SAM HOFFMAN

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Solution-driven mechanical engineering student skilled in rapidly developing and advancing systems with extreme ownership. Collaborative, goal-oriented, nimble problem-solver. Effective communicator with proven track record of owning products from ideation through design, analysis, and build, producing reliable results in professional, real-world settings.

WORK EXPERIENCE

SpaceX – Propulsion Intern, Raptor Chamber and Nozzle *Hawthorne, CA | Summer 2026*

- **Owned the deletion of 70 piece parts** from Raptor Vacuum (saves 13+ hrs of mfg time, reduces engine mass). **Proved feasibility with preliminary calculations and defended to stakeholders.** Developed automated laser welding process.
- **Qualified braze part to buy down risk for flight through hand calculations**, hydro testing. Implemented, flying soon.
- Designed (**NX/ANSYS FEA**) and fabricated tooling and parts to retrofit production assets. **In use for rate production now.**
- **Pitched significant design change to save 30+ hrs mfg time, delete 72 parts, and improve heat flow.** Now in development.
- Re-activated 2 stations after factory move. Consolidated racks to save floor space. Redesigned part to improve weldability.

National Aeronautics and Space Administration – Propulsion Design Intern *Marshall Space Flight Center | Summer 2025*

- **Redesigned propellant ducts** on AM LOX/LNG gas generator. Eliminated 5/11 ports, 2/4 propellants, greatly reducing machining operations, weight, and complexity. Upgraded single OX stream to quadruplet swirl OX injection at spark base.
- **Designed modular fixture system for 1 current and 60+ future satellites** (drafted in **PTC Creo**, validated in **ANSYS FEA**). Used GD&T to guarantee part interfaces. **Developed spacecraft test model** to test integration and fueling procedures.
- Performed strength analysis on intricate rocket nozzle in ANSYS. **Created custom material data, validated max load case.**
- Determined FOS for **high-pressure cryo joint** using gapping/fastener analysis (validated with hand calculations, **ANSYS FEA).**

Honeywell Aerospace – CAES – Mechanical Engineering Intern *Lansdale, PA | Summer 2024*

- **Designed 17 parts** and ~275lb test fixture for precision testing (drafted in **SolidWorks/PTC Creo**, validated in **ANSYS FEA**).
- **Owned and defended design** through 5 in-depth reviews while **saving 60% cost** by tactically tolerancing and sourcing parts.
- Employed Python to expedite large-scale model processing, **creating 1100% time savings** and eliminating repetitive work.
- Created proprietary adapter design for **6400 production units,** redesigned failed chip bracket **to restore $50,000+ server.**

Project Wavefront – Co-Founder & Propulsion Engineer *Philadelphia, PA | April 2025 - Present*

- **Lead the design/build of the world's largest student-designed actively cooled rotating detonation rocket engine** (GOX/NG).
- **Conduct fluidic, thermal, and structural analysis** (**ANSYS**) to influence design and validate engine/test stand through CDR**.**

Penn Jet Propulsion – Propulsion Systems Lead | Mechanical Engineer *Philadelphia, PA | Sep 2023 – June 2025*

- **Led 2024-25 team, optimized variable jet engine exhaust geometry** with Air Force Research Lab under $19,000 contract.
- Created a diverging or converging nozzle for exhaust conditions, **won 2025 Design Award** for inventive, lightweight design.
- **Designed/built 10" turboprop intake, drive housing.** Optimized 110% reverse thrust, **won 2024 Design Creativity Award**

FlashPCB – Production & Systems Testing Intern *Philadelphia, PA | Spring 2023*

- **Designed test PCBs** for 6 areas of production to evaluate AI, **applied Python to scale Eagle/KiCAD files by 500x** for efficiency.

At School: Machine Shop Assistant, Graduate Design & Machining Course TA *Philadelphia, PA | Fall 2024/25 - Present*

EDUCATION

University of Pennsylvania, BSE & MSE in Mechanical Engineering **|** GPA: 3.66 (UG), 4.00 (G)/4.00 *2023 – May 2027 Graduation*

Relevant Coursework: Compressible Flow/Combustion for Rocket Engines, Thermodynamics, Heat/Mass Transfer, Fluid Mechanics, Aerodynamics, Hovering Vehicle Design, CAD & Machining Design, Statics/Dynamics

EXTRACURRICULARS

Penn Engineers Without Borders – Robotics Lead, International Committee CAD Lead *Sep 2023 – Present*

- Designed and built glass vacuum, designed 2 biodigesters to support 8 latrines for Guatemala school.

Stirling Engine (1443 rpm, 61 elements) – Designer **(GD&T, tolerance stacks),** Machinist (18 parts) *Jan 2024 – May 2024*

American Society of Mechanical Engineers – Vice President (University of Pennsylvania Chapter) *April 2024 – February 2026*

Boy Scouts of America – Eagle Scout (160-hr project, Gold and Bronze Palms), Senior Patrol Leader, Instructor *2011 – 2023*

Others: **Bell Society, Hexagon Society, Wharton Aerospace Conference**

SKILLS/INTERESTS

Software: CAD (NX, PTC Creo, SolidWorks) | FEA & CFD (ANSYS) | Microsoft Office | Teamcenter, Windchill, ENOVIA | MasterCAM

Other: Python, MATLAB, ABB RAPID (developing), G-Code | GD&T | 3 axis CNC Mill & CNC Lathe Machining | LATEX

Interests: Rocket/Jet Engines, Space Exploration, Aeronautics, Electric Vehicles, F1, Cooking, Baseball, Skiing, Camping, Hiking